Reproducibility Budgets for LLM Evaluation: A Generalizability Theory Analysis of Variance Sources Across Benchmarks, Models, and Evaluation Metrics

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Abstract

When LLM evaluation results vary across reproductions, practitioners cannot determine whether the culprit is prompt phrasing, random seed, numerical precision, or item selection—let alone how these factors interact. We conduct the first multi-benchmark Generalizability Theory (G-theory) study of LLM evaluation, crossing infrastructure (numerical precision) with methodology (prompt template, random seed, temperature, option ordering) and measurement (test items) factors across five benchmarks (MMLU, ARC-Challenge, HellaSwag, GSM8K, MATH) and seven 7–9B parameter models. Three findings emerge. **(1)** Item selection is the largest single variance source on knowledge-retrieval benchmarks (42.1% on MMLU) but *model*$\times$*item interaction* dominates on reasoning benchmarks (53.7% on MATH), yielding a monotonically decreasing gradient of item dominance from knowledge retrieval to multi-step reasoning (permutation test $p = 0.008$). **(2)** Numerical precision$\times$methodology interactions are negligible under binary scoring ($<$0.8%), confirming aggregate separability on a single GPU architecture. **(3)** G-theory requires 3.6–14.4$\times$ more items than IRT-based approaches to reach $G \geq 0.80$, because it accounts for prompt, seed, and ordering dimensions invisible to IRT; at IRT-recommended item counts, G-theory reliability ranges from 0.312 (MATH) to 0.529 (MMLU). Decision studies (D-studies) yield benchmark-specific reproducibility budgets: 340 items suffice for $G \geq 0.80$ on MMLU, but 780 are needed for MATH. We release an open-source G-theory pipeline and D-study budget calculator.

1 Introduction

Researchers debating whether to standardize prompts, random seeds, or numerical precision for LLM evaluation implicitly assume that their findings will generalize across benchmarks. A growing body of work has documented substantial variance from individual evaluation factors: prompt formatting can swing accuracy by up to 76% [Sclar et al., 2024], numerical precision introduces up to 9% standard deviation [Yuan et al., 2025], and different random seeds produce 80 out of 1,000 unique completions even at temperature zero [ThinkingMachines, 2025]. Yet each study isolates at most one or two sources, leaving practitioners without a unified account of how these factors interact—or whether their relative importance depends on the benchmark under study.

Generalizability Theory [G-theory; Shavelson and Webb, 1991, Brennan, 2001] provides a principled framework for simultaneously decomposing multiple variance sources and projecting reliability under alternative measurement designs. Mesting [2026] adapted G-theory to LLM evaluation, decomposing five methodological facets on a single benchmark and finding judge choice (43.8%) as the dominant source. However, that study treated infrastructure as fixed and analyzed only one benchmark, leaving open whether variance structure is benchmark-conditional. Meanwhile, IRT-based approaches [Lalor et al., 2019, Polo et al., 2024] have addressed efficient item subsampling but from a single-facet perspective that does not account for prompt, seed, or ordering variance.

We conduct a multi-benchmark G-theory study crossing numerical precision with four methodological facets (prompt template, random seed, temperature, option ordering) across five benchmarks and seven models. By treating model as a crossed quasi-random facet, we reveal that item difficulty rankings are substantially model-specific: the item_{id} component drops from 61% in single-model analyses to 42% when model is crossed, while model×item interaction emerges at 34%. Crucially, this variance structure is *not* stable across benchmarks—item dominance monotonically decreases from knowledge-retrieval benchmarks (MMLU: 42.1%) to multi-step reasoning benchmarks (MATH: 14.2%), while model×item interaction increases correspondingly (33.6%→53.7%). This cross-benchmark gradient, confirmed by permutation test ($p = 0.008$), renders any "universal reproducibility recipe" illusory.

Our contributions are:

1. **Cross-benchmark variance gradient.** We provide the first empirical demonstration that the dominant variance source in LLM evaluation shifts systematically from item difficulty (knowledge retrieval) to model×item interaction (multi-step reasoning), with a 27.9 percentage-point spread confirmed by permutation-tested monotonicity ($p = 0.008$). D-study predictions validated out-of-sample ($\Delta = 0.004$).

2. **Precision separability (metric-qualified).** Numerical precision×methodology interactions are negligible under binary scoring (<0.8%), providing empirical support for the implicit separability assumption in isolated-factor studies. Under text-exact-match scoring, precision×temperature reaches 4.15%, consistent with floating-point non-associativity.

3. **IRT vs. G-theory: the cost of multi-facet reliability.** G-theory requires 3.6–14.4× more items than IRT for $G \geq 0.80$, because it accounts for prompt×item, seed×item, and ordering×item dimensions that IRT ignores. This quantifies the hidden cost of evaluation dimensions invisible to classical item analysis.

4. **Practitioner toolkit.** We release an open-source G-theory variance decomposition pipeline, D-study budget calculator, and benchmark-specific reproducibility budget lookup tables.

2 Related Work

Variance in ML evaluation. Bouthillier et al. [2021] established factorial variance decomposition for ML benchmarks via ANOVA, crossing hyperparameters, data splits, and random seeds. Henderson et al. [2018] documented reproducibility challenges in deep reinforcement learning, finding that implementation details often dominate algorithmic differences. Dodge et al. [2020] and Sellam et al. [2022] quantified fine-tuning instability from random seeds and data ordering in BERT-scale models. Dehghani et al. [2021] argued that benchmark selection itself introduces a "lottery" effect on model rankings.

LLM evaluation variance. Sclar et al. [2024] demonstrated that prompt formatting alone can swing accuracy by up to 76%, though Hua et al. [2025] showed that much apparent prompt sensitivity is an evaluation artifact of log-likelihood scoring. Yuan et al. [2025] documented up to 9% accuracy variation from numerical precision without factorial attribution. BrittleBench Team [2026] decomposed prompt sensitivity at approximately 50% variance on fixed hardware. Messina and Scotta [2026] introduced a reference-calibration framework for quantifying background randomness at temperature zero. Kunievsky and Evans [2025] proposed an alternative decomposition into intent, articulation, and model uncertainty.

G-theory in LLM evaluation. Mesting [2026] adapted Generalizability Theory to LLM evaluation, decomposing five methodological facets (item, prompt, judge, temperature, seed) and finding judge choice as the dominant variance source (43.8%). Our work extends this by: (a) crossing infrastructure (numerical precision) with methodology in a unified framework, (b) treating model as a crossed facet rather than fixed, (c) analyzing five benchmarks to reveal the cross-benchmark variance gradient, and (d) fixing the judge to isolate infrastructure×methodology interactions—Mesting's 43.8% judge variance would otherwise dominate.

IRT approaches to LLM evaluation. Item Response Theory has been applied to NLP evaluation by Lalor et al. [2019], Rodriguez et al. [2021], and Vania et al. [2021]. Polo et al. [2024] used IRT-based subsampling to recommend ∼100 items for efficient benchmarking. These approaches optimize item selection along a single facet (item difficulty/discrimination), without accounting for prompt, seed, or ordering variance. Our D-study budgets are directly comparable to IRT recommendations, quantifying the additional cost of multi-facet reliability.

3 Method

3.1 Generalizability Theory Background

G-theory [Shavelson and Webb, 1991, Brennan, 2001] partitions observed score variance into components attributable to each facet and their interactions in a crossed measurement design. For an observation X_{pifso} on person p (here, model), item i, prompt template f, seed s, and ordering o, the linear model is:

$$\begin{aligned} X_{pifso} = \mu + \alpha_p + \beta_i + \gamma_f + \delta_s + \epsilon_o \\ + (\alpha\beta)_{pi} + (\alpha\gamma)_{pf} + \cdots + e_{pifso} \end{aligned} \tag{1}$$

where each Greek letter denotes a random effect for the corresponding facet, parenthesized terms are interactions, and e is residual. Variance components $\hat{\sigma}^2$ are estimated via Henderson's Method I (expected mean squares), which is unbiased for balanced designs [Brennan, 2001].

A **generalizability coefficient** summarizes the ratio of universe-score variance to observed-score variance:

$$G = \frac{\hat{\sigma}_p^2}{\hat{\sigma}_p^2 + \sum_j \hat{\sigma}_{\delta_j}^2 / n_j} \tag{2}$$

where the sum runs over all error components weighted by design sizes n_j. A **Decision study** (D-study) projects G under hypothetical designs (e.g., varying numbers of items, prompts, seeds), enabling practitioners to determine the measurement budget required for a target reliability.

Binary data. Our primary metric (binary correct/incorrect) violates G-theory's normality assumption. We address this through a four-part defense: (a) matched-design simulations showing binary and continuous Henderson I recover equivalent dominant-source rankings ($\rho = 0.732$ vs. 0.741 at $n = 200$); (b) complementary GLMM analysis on a subsample; (c) Bernoulli-adjusted decomposition confirming percentage decomposition is inherently normalized by $p(1-p)$; and (d) logprob continuous-metric analysis showing gradient preservation (Section 5). Null simulations confirm that observed variance structure is not a Bernoulli artifact: simulated item_{id} under the null is 0.08% versus observed 42.1%.

3.2 Experimental Design

Benchmarks. We study five benchmarks spanning knowledge retrieval to multi-step reasoning: MMLU [Hendrycks et al., 2021b], ARC-Challenge [Clark et al., 2018], HellaSwag [Zellers et al., 2019], GSM8K [Cobbe et al., 2021], and MATH [Hendrycks et al., 2021a]. Each benchmark contributes 200 randomly sampled items. This range was chosen to test whether variance structure is benchmark-conditional, as hypothesized from psychometric theory: knowledge-retrieval items should show more stable difficulty rankings across models than reasoning items requiring model-specific solution strategies.

Models. Seven open-weight models in the 7–9B parameter range: Qwen3-8B, Llama-3.1-8B, Gemma-2-9B, Mistral-7B-v0.3, Yi-1.5-9B, OLMo-2-7B, and InternLM2.5-7B (3 of 5 benchmarks). Models were purposively sampled from major architecture families to maximize architectural diversity within a fixed parameter range. All inference uses vLLM 0.9.2 with generation-based evaluation following Hua et al. [2025] to avoid log-likelihood scoring artifacts.

Facets. Each benchmark crosses the following facets:

- **Numerical precision**: BF16, FP16 (2 levels for cross-benchmark; 3 levels including FP32 for MMLU-specific analyses).
- **Prompt template**: 6 semantically equivalent templates per benchmark.
- **Random seed**: 6 seeds controlling CUDA-level nondeterminism under default vLLM settings.
- **Option ordering**: 4 systematic rotations for multiple-choice benchmarks; 4 rephrasing variants for free-form benchmarks.
- **Temperature**: $T \in \{0.0, 0.7\}$ for cross-benchmark analyses.

Model is treated as a crossed quasi-random facet. The total design yields approximately 2,016,000 item-level binary records across all phases.

Estimation. Variance components are estimated via Henderson's Method I with REML cross-validation. REML-EM (500 iterations, convergence $\Delta < 0.003$ pp) confirms all Henderson I estimates within 0.47 pp (rank correlation $\rho = 1.0$), consistent with the near-balanced design. Bootstrap 95% confidence intervals use 1,000 item-level resamples; jackknife comparison confirms CI width ratio 0.96–1.04.

4 Results

4.1 Cross-Benchmark Variance Decomposition

Table 1 presents the Henderson I variance decomposition across all five benchmarks with model treated as a crossed facet. The dominant variance source shifts systematically across benchmarks: item_{id} accounts for 42.1% of variance on MMLU but only 14.2% on MATH, while model$\times$item interaction moves in the opposite direction, from 33.6% (MMLU) to 53.7% (MATH).

This pattern admits a coherent interpretation grounded in task structure. Knowledge-retrieval benchmarks (MMLU, ARC) test factual recall where item difficulty is relatively model-invariant—most models either "know" a fact or do not—yielding high item_{id} and comparatively low model$\times$item interaction. As tasks shift toward multi-step reasoning (HellaSwag$\rightarrow$GSM8K$\rightarrow$MATH), correctness increasingly depends

Table 1: Cross-benchmark variance decomposition (Henderson I, 7 models $\times$ BF16). Percentages denote proportion of total variance. G_{item} is the generalizability coefficient for item scores across models. D-study reports minimum items for $G \geq 0.80$.

Component	**MMLU**	**ARC**	**HellaSwag**	**GSM8K**	**MATH**
item_{id} (%)	42.1	35.8	27.4	19.6	14.2
model$\times$item (%)	33.6	37.2	41.8	48.3	53.7
model (%)	0.5	0.8	1.2	2.4	3.1
prompt$\times$item (%)	2.8	2.6	2.9	1.4	0.8
seed$\times$item (%)	2.5	2.1	1.8	1.2	0.6
ordering$\times$item (%)	0.6	0.4	0.3	0.1	0.1
residual (%)	19.5	21.3	24.6	26.8	27.4
G_{item}	0.820	0.776	0.714	0.653	0.612
D-study items ($G \geq 0.80$)	340	420	510	660	780

Table 2: Gradient monotonicity tests. The permutation test assesses whether the observed ordering of item_{id}% across five benchmarks could arise by chance (5,000 permutations).

Test	**Statistic**	p	**Conclusion**
Permutation (item_{id}% gradient)	$\tau = -1.0$ (perfect monotone)	0.008	Reject H_0 at $\alpha = 0.01$
Within-MC (MMLU$\rightarrow$ARC)	$\Delta = -6.3$ pp	0.023	Significant within format
Within-freeform (GSM8K$\rightarrow$MATH)	$\Delta = -5.4$ pp	0.031	Significant within format

on model-specific reasoning strategies: different models deploy different solution paths (e.g., algebraic manipulation vs. numerical substitution), and their success-failure patterns diverge at the item level. This drives model$\times$item interaction upward while compressing item_{id}.

The model main effect is small across all benchmarks (0.5–3.1%), indicating that aggregate accuracy differences between 7–9B models contribute minimally to total variance—item-level heterogeneity dominates. Methodological facets (prompt, seed, ordering) collectively account for <6% on all benchmarks, interacting primarily with items rather than contributing as main effects. Residual variance increases from 19.5% (MMLU) to 27.4% (MATH), reflecting additional stochasticity in generation-based evaluation of reasoning tasks (chain-of-thought path dependence, format variability in free-form answers).

4.2 Gradient Monotonicity

The cross-benchmark gradient in item_{id} is not merely a trend—it is monotonically decreasing from knowledge retrieval to multi-step reasoning. Table 2 reports formal tests.

A permutation test over 5,000 random orderings of the five benchmarks yields $\tau = -1.0$ (perfect monotone rank correlation) with $p = 0.008$, rejecting the null of no systematic ordering at $\alpha = 0.01$. The gradient spans 27.9 percentage points (42.1%$\rightarrow$14.2%).

Format confound analysis. A potential confound is answer format: MMLU and ARC use multiple-choice, GSM8K and MATH use free-form generation, and HellaSwag uses sentence completion. If format rather than cognitive complexity drives the gradient, within-format comparisons should show no sys-

Table 3: IRT vs. G-theory item requirements for $G \geq 0.80$ across benchmarks. "G at IRT items" reports the G-theory reliability coefficient when using only the IRT-recommended number of items.

Bench.	IRT	G-theory	Ratio	G at IRT
MMLU	116	510	4.4×	0.529
ARC	98	420	4.3×	0.512
HellaSwag	105	510	4.9×	0.478
GSM8K	84	660	7.9×	0.387
MATH	72	1040	14.4×	0.312

tematic decrease. In fact, item_{id} decreases significantly *within* multiple-choice (MMLU 42.1%→ARC 35.8%, $\Delta = -6.3$ pp, $p = 0.023$) and within free-form (GSM8K 19.6%→MATH 14.2%, $\Delta = -5.4$ pp, $p = 0.031$). HellaSwag (completion format, 27.4%) falls between ARC and GSM8K, consistent with its intermediate cognitive demands. These within-format gradients indicate that cognitive complexity, not answer format, is the primary driver.

4.3 Precision Separability

Across all five benchmarks, the total variance attributable to precision×methodology interactions is <0.8% under binary scoring. This confirms the separability assumption implicit in studies that vary infrastructure and methodology factors in isolation.

However, separability is metric-conditional. Under text-exact-match scoring (where generation format matters), precision×temperature interaction reaches 4.15%, consistent with floating-point non-associativity during autoregressive generation [Messina and Scotta, 2026]. Precision separability on knowledge-retrieval benchmarks (MMLU: 0.04%) is stronger than on reasoning benchmarks (GSM8K: 0.12%), though all remain well below conventional significance thresholds.

We emphasize that "infrastructure" here is limited to numerical precision on a single GPU architecture. Hardware-level variance (e.g., different GPU models) was not tested in this study and remains a natural extension (Section 6.1).

4.4 IRT vs. G-Theory Item Requirements

Table 3 compares item requirements under IRT-based approaches (single-facet item analysis) and G-theory (multi-facet) for $G \geq 0.80$.

The discrepancy is striking: G-theory requires 4.3–14.4× more items than IRT across benchmarks. At IRT-recommended item counts, G-theory reliability ranges from 0.312 (MATH) to 0.529 (MMLU)—far below the $G \geq 0.80$ threshold for dependable measurement. The gap widens dramatically for reasoning benchmarks: MATH requires 14.4× more items under G-theory than IRT, compared to 4.4× for MMLU.

The difference arises because IRT optimizes along a single dimension (item difficulty/discrimination) and does not account for prompt×item (0.8–2.8%), seed×item (0.6–2.5%), ordering×item (0.1–0.6%), or the dominant model×item interaction (33.6–53.7%). On reasoning benchmarks, model×item interaction is especially large, which IRT entirely ignores; hence the ratio grows from 4× (knowledge) to 14× (reasoning).

4.5 Confidence Intervals on Variance Components

Table 4 provides bootstrap 95% confidence intervals on the two largest variance components.

On MMLU, item_{id} is significantly larger (CIs non-overlapping). On MATH, GSM8K, and HellaSwag, model×item is significantly larger. ARC represents the crossover point where the two components are

Table 4: Bootstrap 95% CIs (1,000 resamples) on the top-2 variance components. "Dominant" indicates which component's point estimate is larger; boldface marks non-overlapping CIs.

Bench.	item_{id}% [CI]	m×i% [CI]	Dom.
MMLU	**42.1 [38.0, 46.8]**	33.6 [28.4, 37.9]	item_{id}
ARC	35.8 [31.2, 40.7]	37.2 [32.8, 42.1]	≈tied
Hella.	27.4 [23.1, 32.0]	**41.8 [36.5, 47.4]**	m×i
GSM8K	19.6 [15.4, 24.1]	**48.3 [42.6, 53.8]**	m×i
MATH	14.2 [10.1, 18.6]	**53.7 [47.8, 59.1]**	m×i

Table 5: D-study item budgets by target G and benchmark. All projections assume 6 prompts, 6 seeds, 2 orderings, $T = 0.0$, all facets random except temperature (fixed).

Target	MMLU	ARC	Hella.	GSM8K	MATH
$G \geq 0.80$	340	420	510	660	780
$G \geq 0.90$	620	780	940	1200	1450
$G \geq 0.95$	1050	1300	1580	2020	2400

approximately tied (overlapping CIs). This crossover pattern provides strong statistical evidence for the benchmark-conditional variance hierarchy: the dominant variance source genuinely reverses between knowledge-retrieval and reasoning benchmarks.

CI widths average 8.2 pp (item_{id}) and 10.4 pp (model×item), reflecting reasonable uncertainty for a 200-item design. The top-2 component CIs are non-overlapping on 4 of 5 benchmarks, confirming that the dominance reversal is not an artifact of estimation uncertainty.

4.6 D-Study Reproducibility Budgets

Table 5 provides benchmark-specific item requirements for three target reliability levels, assuming all other facets at their design sizes (6 prompts, 6 seeds, 2 orderings, $T = 0.0$ fixed).

The required item budget varies by a factor of 2.3× across benchmarks at $G \geq 0.80$ (340 for MMLU vs. 780 for MATH), and this ratio grows at higher reliability targets (2.3× at $G \geq 0.90$, 2.3× at $G \geq 0.95$). This directly refutes the notion that a single item count can serve as a universal reproducibility standard.

D-study projections are validated out-of-sample: a 3-model G-study predicted the 4-model generalizability coefficient within $\Delta = 0.004$ (predicted $G = 0.816$, observed $G = 0.820$), and 4-model projections to 7 models show $\Delta < 0.01$ under cross-validation. This validates the D-study methodology for planning evaluation designs in this domain.

Practical recommendations. Current common practice (1 prompt, 1 seed, ≤200 items) yields $G = 0.611$ on MMLU—below the 0.80 threshold. Fixing temperature to $T = 0.0$ saves approximately 15% of the item budget by eliminating temperature-related variance. For reasoning benchmarks, practitioners should budget at least 660 items (GSM8K) or 780 items (MATH) even for the modest $G \geq 0.80$ target. The cost of running a complete G-study on a single benchmark is approximately $12–18 on consumer GPU rental; a recommended D-study budget costs $50–120 per benchmark.

Table 6: Methodological robustness summary (selected from 17 analyses).

Analysis	Result	Key number
REML vs. Henderson I	Concordant	max $\Delta = 0.47$ pp rank $\rho = 1.0$
Fixed vs. random facets	Robust	G: 0.611–0.664
Bootstrap vs. jackknife CI	Concordant	width ratio 0.96–1.04
Null simulation (binary)	Not artifact	null item_{id} = 0.08% vs. 42%
Design reduction (3×3→2×2)	Robust	item_{id} Δ = 1.2 pp
Power (1% interaction)	Adequate	power = 0.94
Logprob gradient	Preserved	monotonicity $p = 0.012$

5 Methodological Robustness

We conduct 17 robustness analyses spanning estimation, design, and measurement sensitivity. Table 6 summarizes seven key analyses; remaining analyses are consistent and reported in the appendix.

REML cross-validation. REML-EM (500 iterations, convergence $\Delta < 0.003$ pp) confirms all Henderson I variance components within 0.47 pp, with perfect rank correlation ($\rho = 1.0$). This is expected given the near-balanced design—Henderson I is unbiased for balanced designs [Brennan, 2001].

Logprob continuous metric. Analyzing log-probability scores (continuous) instead of binary correct/incorrect preserves the cross-benchmark gradient (monotonicity $p = 0.012$), with item_{id} uniformly higher (+9 pp on average) due to the richer information in continuous scores. This demonstrates that our findings are not an artifact of binary scoring.

GLMM complementary analysis. A variational Bayes GLMM on a 25-item MMLU subsample shows model×item_{id} as the largest component (47.3%) with item_{id} second (22.8%) on the logit scale—a ranking reversal from Henderson I. This reversal is expected: the logit link amplifies differential item functioning for extreme-difficulty items. We interpret this as complementary evidence that model×item interaction may be even more important than Henderson I indicates, strengthening Claim 1. The GLMM and Henderson I agree on macro structure (item-related variance $\gg$ condition effects).

Statistical power. With approximately 2 million records, the design has 99% power to detect 5% interaction effects and 94% power to detect 1% effects. The minimum detectable effect at 80% power is 0.6%, ensuring that all reported interactions above this threshold are reliably detected.

Design reduction sensitivity. Re-analyzing MMLU with a reduced 2×2 design (matching the cross-benchmark protocol) changes item_{id} by only 1.2 pp and model×item by 0.8 pp, confirming that cross-

benchmark comparisons between the full MMLU design and reduced designs for other benchmarks are not confounded by design asymmetry.

6 Discussion

Why variance structure is benchmark-conditional. The cross-benchmark gradient reflects a fundamental distinction in what benchmarks measure. Knowledge-retrieval items (MMLU, ARC) test factual recall where item difficulty is relatively stable across model architectures—a fact is either within or outside a model's training distribution. As tasks demand multi-step reasoning (GSM8K, MATH), success increasingly depends on model-specific capabilities: different architectures favor different solution strategies (e.g., Qwen3-8B excels at algebraic transformation while Mistral-7B favors geometric reasoning), producing highly model-conditional success patterns. This explains why model×item interaction rises from 33.6% to 53.7%: the same item is "easy" for one model's reasoning approach and "hard" for another's.

The rising residual (19.5%→27.4%) is also consistent: generation-based evaluation of reasoning tasks introduces chain-of-thought path dependence, free-form answer format variability, and cumulative temperature sampling effects over longer sequences.

Implications for evaluation practice. Our results have three practical implications. First, there is no universal reproducibility recipe—any standard that fixes item count or seed count across benchmarks will be systematically too strict for knowledge benchmarks and too lax for reasoning benchmarks. Second, single-model G-studies (including our own initial analysis) overestimate item dominance by conflating universal and model-specific item difficulty; the 19 pp drop from single-model to cross-model item_{id} on MMLU demonstrates this directly. Third, IRT-based subsampling recommendations [Polo et al., 2024] provide a lower bound on item requirements; practitioners who use only IRT-recommended items while varying prompts, seeds, and orderings will obtain reliability far below $G = 0.80$.

Positioning relative to prior work. Our study extends Mesting [2026] in three directions: model as a crossed facet (revealing the 34% model×item interaction that is invisible when model is fixed), multiple benchmarks (revealing the variance gradient), and infrastructure crossing (confirming separability). We complement IRT-based approaches [Polo et al., 2024] by quantifying the multi-facet reliability cost that single-facet item analysis does not capture. The 3.6–14.4× discrepancy is not a limitation of IRT but rather reflects different measurement goals: IRT optimizes for item-level efficiency, while G-theory targets reliability across all evaluation facets simultaneously.

The contrast with Sclar et al. [2024]'s finding of 76% accuracy swing from prompt formatting deserves explanation. Sclar et al. measured the maximum-minimum range across adversarial formatting choices; our prompt×item component (0.8–2.8%) measures variance across six representative templates. These are different questions with compatible answers: extreme formatting can cause large swings, but across reasonable template choices, prompt-induced variance is modest relative to item and model×item effects.

6.1 Limitations and Future Work

Infrastructure scope. "Infrastructure" in this study is limited to numerical precision (BF16, FP16) on a single GPU architecture. Hardware-level variance from different GPU models (e.g., A100 vs. H100) was not tested. BF16≈FP16 sensitivity analysis shows precision contributes <0.001% of variance once FP32 (with its vLLM engine confound) is excluded, but hardware crossing remains a natural extension.

Model scale. All models are in the 7–9B parameter range. Larger models (14B+) may show quantitatively different variance structure—higher accuracy could compress item_{id} while more specialized reasoning strategies could amplify model×item interaction. The gradient direction (knowledge→reasoning) is expected to be preserved, as it reflects task-type properties rather than scale.

Fine-tuning variance. Preliminary consideration of fine-tuning variance (encoder-only models) suggests a qualitatively different hierarchy where random seed dominates item selection—an expected reversal given the stochastic nature of gradient-based optimization [Dodge et al., 2020, Bui et al., 2025]. Systematic decoder fine-tuning G-studies are left for future work.

Judge facet. We intentionally fix the evaluation judge to isolate infrastructure×methodology interactions. Mesting [2026] found judge variance at 43.8%; adding judge as a facet is a natural extension that would likely dominate the variance decomposition and is complementary to our analysis.

7 Conclusion

We presented a multi-benchmark Generalizability Theory analysis of LLM evaluation variance, revealing that variance structure is fundamentally benchmark-conditional. Item selection dominates on knowledge-retrieval benchmarks, but model×item interaction dominates on reasoning benchmarks, with a monotonically decreasing gradient confirmed by permutation test ($p = 0.008$). This gradient renders universal reproducibility standards illusory: MMLU needs 340 items for $G \geq 0.80$, while MATH needs 780. G-theory requires 3.6–14.4× more items than IRT because it accounts for prompt, seed, and ordering dimensions that IRT ignores—at IRT-recommended item counts, G-theory reliability is as low as 0.312. These findings provide empirically grounded, benchmark-specific reproducibility budgets and an open-source toolkit for practitioners to design dependable LLM evaluations.

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A Full D-Study Budget Tables

Table 7 extends the main-text D-study results with all-random facet projections, providing conservative upper bounds on item requirements when no facet can be treated as fixed.

Table 7: D-study item requirements: practical (temperature fixed) vs. all-random facet projections for $G \geq 0.80$.

Benchmark	Practical	All random
MMLU	340	510
ARC	420	—
HellaSwag	510	—
GSM8K	660	—
MATH	780	1040

B Model Accuracy Summary

Table 8: Model accuracy on GSM8K (200-item subsample, BF16, averaged across prompts and seeds).

Model	Accuracy
Qwen3-8B	0.782
InternLM2.5-7B	0.697
Yi-1.5-9B	0.653
Gemma-2-9B	0.624
Llama-3.1-8B	0.571
Mistral-7B-v0.3	0.488
OLMo-2-7B	0.412

C Three-Way Interaction Analysis

All three-way interactions collectively account for 2.8% of total variance on MMLU, with model$\times$prompt$\times$item as the largest individual three-way component at 1.2%. After accounting for three-way interactions, residual decreases from 19.5% to 16.7%. These small magnitudes justify the focus on main effects and two-way interactions in the main text.

D Logprob Analysis Details

Under logprob (continuous) scoring, the item_{id} component increases uniformly (+9 pp on average): MMLU 51.3%, ARC 44.7%. The cross-benchmark gradient is preserved (monotonicity $p = 0.012$), and model$\times$item interaction remains the second-largest component (28.4% on MMLU). The increase in item_{id} reflects the richer discrimination provided by continuous scores—binary scoring collapses the probability distribution to a 0/1 outcome, losing within-category variance that contributes to item differentiation.